

CO2-AT-1

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ANALYTICAL PROBLEM SOLVING.

Q1. Linked List Configuration

L1 = Empty

L2 = Empty

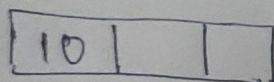
a) operation

L1.append(10)

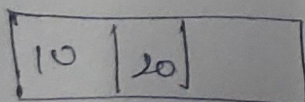
L1.append(20)

L1.append(15)

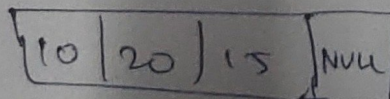
append(10)



After append(20)



After append(15)



b) operation

L2.append(10)

L2.append(20)

L2.append(15)

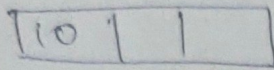
L2.moveToStart()

L2.insert(30)

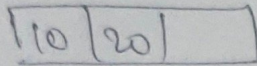
L2.next()

L2.insert(12)

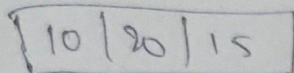
Append (10)



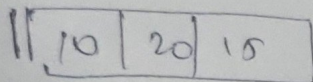
Append (20)



Append (15)

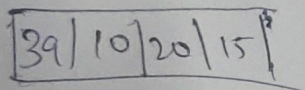


After Move to Start()

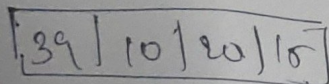


↑
current

After insert(29)

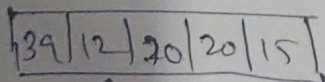


After next()



↑
current

After insert(12)



↑

current position 12.

2. Binary Search.

Given elements.

66 111 212 323 432 543 643 649 777 888

index	0	1	2	3	4	5	6	7	8	9
value	66	111	212	323	432	543	643	649	777	888

Case 1: key 111

low = 0

high = 9

mid = $9/2 = 4$

Element = 432

$111 < 432$

Search left half

low = 0

high = 3

mid = 1

Element = 111

Found at index 1.

Case 2: key = 888

low = 0

high = 9

mid = 4

432

$888 > 432$

Search right

low = 5

high = 9

mid = 7

649

888 > 649

Search right

low = 8

high = 9

mid = 8

777

888 > 777

Search right

low = 9

high = 9

mid = 9

888 found at index 9.

Case 3: key 999

low = 0

high = 9

mid = 4 → 432

Search right

low = 5

high = 9

mid 7 = 649

Search right

mid 8 = 777

Search right

mid = 9 → 888

Element not found.

Q3

Stack (LIFO)

Array size = 5

operation:

Push(45)

Push(22)

Push(11)

Push(77)

Pop()

Push(44)

Push(33)

Pop()

Pop()

Pop()

Push(45) TOP = 0

45				
----	--	--	--	--

Push(22) TOP = 1

45	22			
----	----	--	--	--

Push(11) TOP = 2

45	22	11		
----	----	----	--	--

Push(77) TOP = 3

45	22	11	77	
----	----	----	----	--

Pop() 77 remove TOP = 2

45	22	11		
----	----	----	--	--

Push(44)

45	22	11	44	
----	----	----	----	--

 TOP 3

Push(33),

45	22	11	44	33
----	----	----	----	----

 TOP = 4

Stack full

Pop()

remove 33

45	22	11	44
----	----	----	----

 TOP = 3

Pop()

remove 44

45	22	11
----	----	----

 TOP = 2

Pop()

remove 11 TOP = 1

45	22
----	----

TOP Element = 22

TOP index = 1

4. Prefix & Postfix.

Expression:

$(a+b)+c*(d+e+f)*(g+h)$

Prefix conversion.

$(a+b)$

Prefix + ab

$(d+e)$

Prefix + de

$c^*(dte)$

$*c+de$

$(a+bc)^*(dte)$

$++ab^*c+de$

$((a+b)+c^*(dte))+f$

$++ab^*c+de$

$+++ab^*c+de\ f$

$(g+h)$

$+gh$

Prefix: $*+++ab^*c+de++gh$

Postfix: $ab+cdet^*+f+gh+^*$

Q5. Queue vs Priority Queue

Simple Queue

- Follows FIFO (First in First out)
- Customers are served in the order they arrive
- Fair because everyone waits their turn
- Enqueue: $O(1)$
- Dequeue: $O(1)$

Arrival:

$A \rightarrow B \rightarrow C \rightarrow D$

Service

$A \rightarrow B \rightarrow C \rightarrow D$

Priority Queue

- Customer are served based on Priority
- VIP customer can move ahead of normal customer
- Normal customer may wait longer
- Fairness is reduced but urgent customer receive faster service.

insert: $O(\log n)$

Delete highest priority: $O(\log n)$

Arrival:

A

B

VIP C

D

Service order:

VIP C

A

B

D